

# Dielectric Rope & Rigging Kit

## Cost Analysis per Unit — CSMFAB 017

**Category:** Aegis Home — Hardware **Unit Basis:** 50m rope + 6 hardware pieces **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

### 1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

### 2. Bill of Materials (BOM)

| Component                                           | Quantity/Unit | Unit Price  | Extended Cost   |
|-----------------------------------------------------|---------------|-------------|-----------------|
| UHMWPE bio-based rope (50m, 12mm)                   | 4.8 kg        | \$15.00/kg  | \$72.00         |
| ZrO2 ceramic thimbles (3pc)                         | 0.3 kg        | \$90.00/kg  | \$27.00         |
| PEEK CF40 shackles (3pc)                            | 0.4 kg        | \$280.00/kg | \$112.00        |
| YInMn Blue UV-stabilized rope jacket (outer sleeve) | 0.5 kg        | \$75.00/kg  | \$37.50         |
| BFRP core braided reinforcement                     | 0.8 kg        | \$22.00/kg  | \$17.60         |
| <b>BOM Subtotal</b>                                 |               |             | <b>\$266.10</b> |

### 3. Labor, Process & Overhead Costs

| Cost Element              | Prototype | Small Batch | Production Volume |
|---------------------------|-----------|-------------|-------------------|
| BOM Materials             | \$266.10  | \$218.20    | \$172.97          |
| Direct Labor              | \$80.00   | \$55.00     | \$35.00           |
| Overhead (15% of labor)   | \$12.00   | \$8.25      | \$5.25            |
| Quality Control & Testing | \$9.60    | \$4.40      | \$1.75            |
| Packaging & Logistics     | \$25.00   | \$20.00     | \$15.00           |

**Tooling & Equipment Notes:** Standard rope braiding equipment — no exotic tooling required.

### 4. Per-Unit Cost Summary

| Scale                         | Total COGS   | Suggested MSRP Range | Gross Margin |
|-------------------------------|--------------|----------------------|--------------|
| Prototype (1-5 units)         | <b>\$520</b> | \$832 – \$1,144      | 35-55%       |
| Small Batch (100-500/yr)      | <b>\$360</b> | \$540 – \$720        | 33-50%       |
| Production Volume (5K-50K/yr) | <b>\$265</b> | \$371 – \$477        | 28-45%       |

**Market Reference:** \$320-480 (safety rope market; dielectric rope is standard for electrical work — existing market)

## 5. Key Cost Drivers (V2.0)

| Driver                    | Impact                                           | Mitigation                                                                      |
|---------------------------|--------------------------------------------------|---------------------------------------------------------------------------------|
| ZrB2/Si3N4 UHTC ceramics  | HIGH — specialty powder \$350+/kg                | Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)               |
| MXene Ti3C2Tx             | VERY HIGH — \$2,000/kg at 2026 scale             | Tile pattern minimizes usage; price projected -60% by 2028 as production scales |
| Elium® thermoplastic BFRP | MEDIUM — recyclability adds value; \$12/kg resin | Phoenix Protocol recycling recovers 100% monomer; net cost offset               |
| Tooling NRE               | HIGH for first production run                    | Paid by pre-orders or grant funding; amortized quickly                          |
| Flash Sintering process   | MEDIUM — shares SPS equipment                    | Regional ceramic sintering facility sharing model                               |

## 6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** \$47 – \$93 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

*End of Dielectric Rope & Rigging Kit Cost Analysis | CSMFAB 017 | June 2026*